Architectures of Absence: AI Governance under the FS AI RMF

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Abstract

The Financial Services AI Risk Management Framework, released by the US Treasury in February 2026, specifies 230 control objectives spanning governance, data, model development, validation, monitoring, third-party risk, and consumer protection. Read as a checklist, the framework is operationally daunting and conceptually opaque; community banks below the primary applicability thresholds face an unresolved question about what commensurate compliance means for their context. We argue that the framework has an underlying coherence — empty-chair representation — that makes both interpretation and operationalization tractable: nearly every control objective can be understood as protecting the interests of a party who is not in the room when an AI system makes or supports a financial decision.

We develop the empty-chair frame at three levels. We argue that AI applications in regulated decision contexts divide into structurally distinct categories distinguished by whether a verification regime is available for their outputs, and that two categorical positions follow: post-hoc explanation is structurally inadequate as a verification regime in adversarial contexts, and human oversight of AI-generated decisions is not a verification regime when the human reviews explanations rather than reasoning. We propose three diagnostic techniques — empty-chair enumeration, pages missing from the ledger, and a produced-silence question pair — that operationalize the frame as architectural design discipline and examination methodology. We propose four categories of architectural primitive that empty-chair representation requires: tamper-evident temporal capture, evidence binding, three-dimensional confidence characterization, and structural drift typology.

Across these arguments, a unifying pattern emerges: the architectures the FS AI RMF was written to govern fail not because their controls are corrupted versions of correct measurements but because their controls are produced to occupy the rhetorical space of accountability while the underlying processes are architecturally suppressed. The frame's contribution is an analytical and architectural vocabulary for surfacing this distinction, with implications for examination practice, architectural design, and the regulatory direction of FS AI RMF operationalization. This is the first paper in a planned sequence; companion papers treat architectural specification at implementation level and examination methodology in their respective registers.

1 Opening Movement

The US Treasury's Financial Services AI Risk Management Framework (FS AI RMF), released in February 2026, specifies 230 control objectives spanning governance, data, model development, validation, monitoring, third-party risk, and consumer protection (U.S. Department of the Treasury 2026). Read as a checklist, the framework is operationally daunting and conceptually opaque — community banks under the threshold for primary applicability still face the question of what *commensurate* compliance means, and the document itself does not answer it. Read as a structure,

however, the framework has an underlying coherence that makes both interpretation and operationalization tractable: nearly every control objective can be understood as protecting an interest of a party who is not in the room when an AI system makes or supports a financial decision.

Recent commentary on the FS AI RMF describes it as a *systems blueprint* rather than a checklist, and urges institutions to map controls to architecture, ownership, and evidence rather than to documentation alone. We extend that move one step further: the blueprint has a normative structure, and that structure is *empty-chair representation* — the design of AI governance such that specific absent parties are structurally represented in the architecture of decision-making, not merely in its documentation.

The party in the empty chair varies by control. In some, it is the loan applicant who will be denied without recourse. In others, the depositor whose funds are deployed in ways they did not consent to. In others, the future compliance officer who will inherit responsibility for decisions made under regimes they did not construct. In others, the examiner applying updated model-risk guidance that had not yet been issued when the system was deployed. In others, the institution itself in its future state, after personnel turnover and ownership changes. The empty chair is plural, and the chairs do not always agree. But the architectural function of regulatory governance — and of any honest AI governance framework — is to make these absent parties audible to the parties present at the moment of decision.

This reframe matters operationally. By *verification regime* we mean the set of technical, procedural, and evidentiary mechanisms by which parties not present at the moment of decision can later assess whether their interests were represented in the reasoning, not merely in the outcome. A control *represents* an absent party only if it induces observable, auditable constraints on system behavior or evidence production that a later examiner could use to assess whether that party's interests were respected. Representation in this sense is architectural; it is not satisfied by documentation that asserts representation occurred.

Examiners reviewing AI deployments at community banks face a problem the regulation does not solve for them: 230 control objectives without an interpretive frame collapse into compliance theater, where institutions demonstrate adherence to controls without demonstrating the underlying integrity the controls were written to ensure. With an empty-chair frame, examination becomes a structured inquiry: *which absent parties does this AI system affect, and what makes their interests audible in the architecture?* Controls that protect those interests have substance; controls that do not are decoration. The frame gives examiners diagnostic leverage the checklist does not.

Two categorical positions follow from the frame, and we mark them explicitly because both are positions the prevailing vendor discourse is structurally incentivized to obscure.

First, *post-hoc explanation is structurally inadequate for high-stakes AI decisions.* This is not a novel claim — Rudin (2019) made it forcefully, and subsequent work has demonstrated that explanation methods can be manipulated (Slack et al. 2020), that chain-of-thought reasoning often does not reflect the model's actual reasoning (Lanham et al. 2023; Turpin et al. 2023), and that text-channel observation systematically fails to verify the epistemic state it appears to expose. Bordt et al. (2022) sharpen the claim with a contextual distinction directly relevant here: post-hoc explanation methods fail specifically in *adversarial* contexts, where the explanation's recipient cannot assume cooperative intent from the system being explained. Bank examination is structurally adversarial — the examiner is testing the institution's claims, not collaborating with the institution to understand its model. The implication for AI governance: any architecture that relies on explanation-of-output as its verification regime in an adversarial context has built a Potemkin form of accountability. The empty chair — the party who depends on the verification regime to represent their interest — is not represented by an explanation that does not bind to the underlying reasoning.

Second, *human oversight of AI recommendations is not a verification regime when the human*

reviews explanations rather than reasoning. This follows from the first but deserves separate naming because the dishonest middle is so commercially attractive. A system in which AI generates a recommendation, a human reviews the explanation, and the human signs the decision satisfies an org-chart picture of accountability without satisfying the verification requirement the org-chart was meant to encode. The signature attests to the human's review of the explanation. The decision was made by the AI. The reasoning was never verified. The empty chair has been moved further from the room, not closer to it.

These categoricals are not a counsel of despair about AI in banking. They are a categorical line between AI uses where verification is structurally available and AI uses where it is not. Pattern recognition across decision histories, anomaly detection in transaction monitoring, surfacing of inconsistencies for human investigation — these are AI applications where the verification regime is concrete: the human investigates the flagged case with full context, and the AI's role is bounded to surfacing rather than deciding. AI applications that generate decision recommendations for human signature operate in a different regime, one where the structural inadequacy of text-channel verification has not been solved and where claims of human oversight do not address the underlying gap. The frame we develop treats these as architecturally different categories, not as gradations of risk along a single axis.

A note on what the empty-chair frame is doing structurally, before the body sections develop it. Absence in regulated decision contexts is not a static gap. Architectures actively produce silences and read those silences as endorsements: the false-positive customer who absorbs the cost of being flagged because appeal is prohibitively expensive; the documentation regime that captures only post-hoc justification and reads the absence of contradicting evidence as fidelity; the compliance pass-rate that is sustained by the institution's success at producing the silences the rate appears to measure against. The frame's diagnostic move is not only *who is absent* but *what is the architecture doing that creates this absence, and what is it inferring from the silence it created?* This second move sharpens the categoricals above and is developed in Section 4 as the operational core of the diagnostic techniques. We mark it here so the reader recognizes the frame's structural commitment from the outset: silence is not evidence of absence when an architecture is producing the silence.

The remainder of this paper develops the empty-chair frame as a methodological tool, examines the architectural primitives that make absent-party representation honest rather than rhetorical, and proposes a structure for AI governance in community banking that aligns with FS AI RMF's underlying intent rather than merely its surface form. We treat the FS AI RMF's risk-and-control matrix and Control Objective Reference Guide as a design space for empty-chair representation, mapping absent-party interests to specific architectural primitives and evidence artifacts. We argue that a framework grounded in this principle is more tractable for community banks than the 230-control-objective surface, more defensible to examiners than checklist-aligned compliance, and more honest about what AI can and cannot do in regulated decision contexts than current vendor approaches admit.

2 The Frame as Method

The empty-chair frame's value depends on its generativity. If naming absent-party representation as the organizing principle behind the FS AI RMF produces only a more sympathetic vocabulary for compliance theater, the frame has earned nothing. If it produces interpretive leverage on specific controls — vocabulary for talking about what a control is for, criteria for assessing whether an implementation represents the parties the control was written to protect, diagnostic moves available to examiners reviewing the implementation — the frame has earned its position as an organizing

principle. This section demonstrates the latter through a worked example in BSA/AML transaction monitoring, and in the course of doing so refines the frame itself by distinguishing two kinds of absence.

We choose BSA/AML rather than lending or underwriting deliberately. Lending is where the empty-chair frame is most obvious — the loan applicant is concrete, the harm is concrete, fair-lending controls have decades of regulatory history to draw on, and the absent party's interest is well-rehearsed. The frame applied to lending tends to confirm what readers already believe, which is a poor test of whether the frame is doing work or merely renaming work that was already being done. BSA/AML is harder. The empty chairs are plural and conflicting in ways lending is not. The institution's interest is more aligned with the regulator's than in lending. The customer's interest is ambiguous — protection from fraud and protection from false flagging point in opposite operational directions. If the frame produces clarity here, where the chairs disagree and the controls do not align cleanly with any single absent party, the frame is doing more than renaming.

2.1 Two kinds of absence

Before enumerating the empty chairs in BSA/AML, we refine the frame. The opening movement treated absence as a single category — parties not in the room when a decision is made. In practice, absence has two structurally distinct forms, and the diagnostic moves available to address them differ.

Structural absence describes parties whose absence is given by the situation: parties who do not yet exist, who have no embodied agent, or whose absence is a designed feature of the regulatory regime. The future compliance officer, the not-yet-victimized customer, the financial system as such, the criminal we are designing to exclude — these are absent because the architecture of the situation places them outside the decision context. The frame's diagnostic move on structural absence is *how does the architecture make this party audible to the parties who are present?*

Produced absence describes parties whose absence is created by the architecture itself, then read by the architecture as evidence of correctness. The customer who would have appealed if appeal were not prohibitively expensive. The borrower who stopped applying after repeated denial and whose absence from the application data is read as evidence of self-selection out of credit demand. The reviewer whose actual reasoning is structurally erased by a documentation regime that captures only post-hoc justification, where the absence of contradicting evidence in the record is read as evidence the documentation is faithful. In each case, the architecture has costs or design choices that suppress a voice, and the architecture then interprets the suppressed voice's silence as endorsement of the decision that suppressed it.

The distinction matters because produced absence is invisible to the empty-chair frame as initially stated. *Make the absent party audible* is the wrong operation when the architecture is producing the silence — the chair appears empty because the architecture removed it, and asking how to give voice to the chair while leaving the removal-mechanism intact will fail. The diagnostic move on produced absence is *what is the architecture doing that creates this silence, and what is it inferring from the silence it created?*

This distinction sharpens position one of the paper. Compliance theater is partly the failure to represent structural absences. But compliance theater's persistence is more substantially about produced absences — about architectures that suppress challenge, then read the absence of challenge as evidence the architecture is correct. The dishonest middle is not merely a failure of representation. It is an active production of silence, interpreted as endorsement.

2.2 The empty chairs in BSA/AML monitoring

Six absent parties have interests in how a community bank monitors transactions for suspicious activity (Federal Financial Institutions Examination Council 2020). Each is structural, produced, or both. We mark which.

The next victim of fraud whose pattern this transaction is part of. Structural absence. When a transaction is part of an emerging fraud pattern, the institution's monitoring decision affects not the customer making the transaction but the customer who will be victimized in the next iteration. The next victim is fully absent from the transaction record and from the institution's customer relationship by the structure of time and the limits of prediction. They are represented in the monitoring architecture or they are not represented at all.

The customer whose account is frozen on a false positive. Mixed: structural and produced. The customer is present in the transaction record but absent from the decision context that determined how their pattern was scored — that absence is structural, given that the model evaluates the transaction without consulting the customer. But the customer's silence after the freeze is partly produced: appeal processes are typically opaque, expensive, and slow, and many false-positive customers absorb the cost rather than contest it. The institution's apparent flagging accuracy is sustained in part by appeal-cost dynamics that suppress challenge from the falsely flagged.

The marginalized customer whose transaction patterns are flagged at elevated rates. Mixed: structural and produced. Models trained on historical SAR data are at risk of inheriting historical flagging biases, especially where enforcement and reporting have differed across communities even when underlying risk has not (U.S. Department of the Treasury 2023). Communities whose transaction patterns differ from the model's training distribution — recent immigrants using money-transfer services, cash-economy participants, members of religious or cultural communities with non-standard remittance patterns — can experience flagging rates that do not track underlying risk consistently across populations. The training-distribution gap is structural absence: the model was not trained to represent these populations as separate. But the appeal-cost dynamic is produced absence: the population most likely to be falsely flagged is also the population least equipped to contest the flag, and the institution's stable false-positive rate across populations is partly an artifact of differential appeal costs rather than evidence of unbiased flagging.

The institution itself, in the form of its future enforcement exposure. Structural absence. The bank that fails to file required SARs faces regulatory action; the bank that over-files faces examiner skepticism about its risk calibration. Both are forms of enforcement exposure that compound over time. The institution's future self — the bank in five years, when current monitoring decisions become the subject of an examination — is an empty chair the present institution can either represent through architectural attestation or fail to represent through reliance on reconstructed reasoning.

The financial system, in the form of monitoring quality across institutions. Structural absence. BSA/AML is a coordinated regulatory regime; one institution's monitoring inadequacy creates externalities for institutions whose monitoring is more rigorous, because illicit activity predictably gravitates toward weaker control environments. The system as a whole is an empty chair represented by inter-agency information sharing, FinCEN reporting, and cross-institution examination practice. The community bank's monitoring architecture either contributes to this representation or free-rides on it.

The criminal whose laundering depends on monitoring inadequacy. Structural absence by intentional design. We name this empty chair explicitly because its absence in the architecture is intentional and structural, and because conflating "we should not represent the criminal" with "we should not consider the criminal's interest in our threat model" is a category error. The criminal's interest is the threat the architecture is designed against. The frame requires naming the chair

we are designing to exclude. We also note the boundary case: the criminal and the marginalized customer can be hard to distinguish from behavior patterns alone, and architectures that conflate the two by design produce a particularly vicious form of produced absence — the marginalized customer flagged as criminal, whose appeal is expensive and whose silence after flagging is read as confirmation of suspicion.

The 230 control objectives do not talk about controls this way. They specify what the institution must do — record-keeping, suspicious-activity reporting, customer due diligence, transaction monitoring thresholds — without naming whose interest each requirement structurally protects, and without distinguishing structural from produced absence. The empty-chair frame's first contribution is the vocabulary that makes both kinds of structure visible: each control's substance is determined by which empty chairs it represents and which it does not, and by whether the absences it leaves unrepresented are structural or produced by the control's own design.

2.3 Walking three controls through the frame

Consider three FS AI RMF control objectives that touch transaction monitoring in different ways. We do not specify the exact control numbers — the framework's structure is more stable than its enumeration, and tying the analysis to specific numbers makes this section brittle. The substantive content is what matters.

Control: monitoring system performance is continuously assessed for drift in detection rates and false-positive rates, with documented thresholds triggering revalidation.

The empty-chair reading: this control protects the institution's future self (drift detection prevents enforcement exposure from gradually degraded monitoring), protects the next victim of fraud (drift detection ensures detection rates remain adequate), and partially protects the marginalized customer (false-positive drift surfaces disparate impact over time, if the false-positive rate is decomposed by population rather than aggregated). Implementation choices determine which empty chair is actually represented. A system that monitors aggregate false-positive rates without decomposition by population represents the institution and the next victim but does not represent the marginalized customer; the false-positive rate can remain stable in aggregate while shifting in distribution across populations. The control's text is satisfied; the marginalized customer's chair remains empty. An implementation that monitors decomposed false-positive rates and triggers revalidation on disparate-impact drift represents the marginalized customer; the implementation choice is the difference between a control that does its work and a control that performs compliance.

A further produced-absence consideration: the false-positive rate the system measures is itself partly an artifact of differential appeal rates. Customers who challenge false flags reduce the measured false-positive count; customers who absorb the cost of being flagged do not. A monitoring system that treats its measured false-positive rate as ground truth is reading the produced silence of customers who could not afford to appeal as evidence of correct flagging. An implementation that distinguishes contested from uncontested flags, and treats the uncontested-flag rate as evidence about appeal costs rather than as evidence about flagging accuracy, represents the produced-absence customer that the aggregate measure suppresses.

This is the frame's first interpretive move: the control's text underspecifies which empty chairs it represents, and the implementation choice is where the substantive question lives. Asking *which empty chair does this implementation represent* is more diagnostic than asking *does this implementation comply with the control text*. Asking *what silence does the implementation produce, and what does it infer from that silence* sharpens the diagnostic further.

Control: AI components used in transaction monitoring have documented inventory entries including model purpose, training data provenance, and validation methodology.

The empty-chair reading: this control protects the institution's future self (an examiner arriving in three years can reconstruct what the system was), protects the future compliance officer (institutional knowledge survives personnel turnover), and indirectly protects the marginalized customer (training data provenance is the precondition for assessing whether the model represents their population adequately). The control does not directly protect the false-positive customer or the next victim; their interests are downstream of model performance, which the inventory does not directly assess.

The implementation question is what counts as adequate documentation. A model card listing high-level capabilities and a training-data summary protects no empty chair effectively — it produces a documented inventory entry without producing the institutional self-knowledge the inventory is for. An entry that captures specific decisions during model development, the empty chairs considered in those decisions, the trade-offs made, and the institutional context in which the model was approved produces inventory that the future compliance officer can actually use. The first is theater; the second is representation. Both satisfy the control text. The frame produces the distinction the text does not.

Control: human review processes for AI-flagged transactions include documentation of reviewer reasoning sufficient for later examination.

The empty-chair reading: this control reaches for the dishonest middle, and produces a particularly clean example of produced absence. The text frames human review as the verification regime; the reviewer's documented reasoning is treated as the audit trail. The empty chairs the control is meant to protect — the false-positive customer, the next victim, the institution's future self — are represented only if the documented reasoning actually captures what the reviewer did. This is where the second categorical from the opening becomes operational: documented reasoning is not verification of reasoning unless the documentation binds to what the reviewer actually thought, and there is extensive evidence from cognitive psychology that retrospective justifications often diverge from the reasoning that actually produced a decision, especially under time pressure and high case volume (Nisbett and Wilson 1977).

The produced-absence dimension: a documentation regime that asks reviewers to write justification after deciding produces narrative that satisfies the control text. The reviewer's actual contemporaneous reasoning — the considerations weighed, the alternatives entertained, the moments of uncertainty — is structurally erased, because the documentation regime did not capture it as it occurred. The institution then reads the absence of contradicting evidence in the documentation record as evidence the documentation is faithful. The reviewer's actual reasoning has been suppressed by the documentation regime, and its silence has been interpreted as endorsement of the regime's faithfulness. This is produced absence at the architectural level, and the control as commonly implemented is one of its principal mechanisms.

A review interface that captures decision context as the reviewer engages with it — what was consulted, what alternatives were considered, what made the difference — produces contemporaneous evidence that does not require the reviewer's silence to be interpreted as endorsement. The control underspecifies; the frame makes the underspecification visible; the produced-absence distinction makes visible what the underspecification costs.

2.4 What the frame produced

Four moves, in this section, the checklist did not produce. First, an explicit accounting of which absent parties have interests in BSA/AML monitoring, including the chair we are designing to exclude. Second, a distinction between structural and produced absence, with different diagnostic moves for each. Third, a vocabulary for distinguishing controls by which empty chairs their implementations represent and which kinds of absence they leave unaddressed, separating control

text from implementation substance. Fourth, a diagnostic question pair — *which empty chair does this implementation represent, and what silence does it produce that it then reads as endorsement?* — that produces leverage on assessment that the compliance question — *does this implementation satisfy the text* — does not.

These are not solutions to the problems the controls address. They are a frame for thinking about the controls that produces interpretive and diagnostic moves the framework's checklist surface does not provide. The frame's claim to organizing principle status rests on whether moves of this kind generalize across controls and across regulated decision contexts, which the remainder of this paper develops. What this section establishes is that the moves exist — that the frame produces structure where the checklist produces only enumeration, and that the structural-versus-produced distinction adds a dimension to the frame that is invisible without it.

3 Verification Regimes as Taxonomic Move

Section 2 demonstrated that the empty-chair frame produces interpretive structure where the FS AI RMF's checklist surface produces only enumeration. This section makes a stronger move: it argues that AI applications in regulated decision contexts divide into two structurally different categories, distinguished not by risk magnitude but by whether a verification regime is *available* for the application's outputs. The categoricals from the opening — post-hoc explanation as structurally inadequate, human oversight as not a verification regime when reviewing explanations — are taxonomic markers rather than empirical observations. They identify where AI applications cross a structural line, after which the verification problem has not been solved by any deployed architecture we know of.

The worked example is consumer lending — specifically, the adverse action notice regime under the Equal Credit Opportunity Act and Regulation B. We choose lending because the dishonest middle is concentrated in a single artifact (the adverse action notice) whose regulatory function is to communicate the reasons for denial to a customer who is structurally absent from the decision but legally entitled to its reasoning. The artifact is a verification-regime claim made to a specific empty chair. Its current form does not survive examination.

3.1 The structural line

Recall the opening's framing: a verification regime is the set of mechanisms by which parties not present at the moment of decision can later assess whether their interests were represented in the reasoning, not merely in the outcome. The categoricals identify two failure modes for verification regimes built on text-channel observation:

First, post-hoc explanation methods do not bind to the underlying reasoning of the model. The explanation is generated by a separate process — feature attribution, counterfactual generation, attention visualization, natural-language summarization — that produces text correlated with the model's output but not constituting the model's reasoning (Rudin 2019; Bordt et al. 2022). The empty chair receives an artifact that resembles reasoning without being reasoning. Verification of that artifact is not verification of the model.

Second, human review of an explanation is review of the artifact, not the underlying process. A reviewer who reads an adverse action notice and signs it has verified that the notice is well-formed, that its reasons are plausible, that no obvious red flags appear. The reviewer has not verified that the reasons given are the reasons the model used, because the reasons the model used are not present in the artifact being reviewed (Lanham et al. 2023; Turpin et al. 2023).

Together these foreclose a category of architectural claim. An AI application that generates a decision recommendation, attaches a post-hoc explanation, and is signed by a human reviewer does not have a verification regime for the empty chair the explanation is addressed to. The explanation cannot be verified against the model's reasoning, and the reviewer's signature attests only to review of the explanation. The empty chair receives an artifact and a signature. The empty chair has not received verification.

This is a structural foreclosure, not a quantitative one. No amount of additional human review, additional explanation detail, or additional documentation closes the gap, because the gap is in the binding between explanation and reasoning, which the architecture does not provide. Until an architecture provides that binding, the application is on one side of the structural line. Once an architecture provides it — through tensor-channel observation, through ante-hoc interpretable models, through other mechanisms whose specification is the work of follow-on papers — the application is on the other side.

3.2 Adverse action notices as the canonical case

The Equal Credit Opportunity Act requires creditors to provide adverse action notices to applicants who are denied credit, stating the specific reasons for the denial (Consumer Financial Protection Bureau 2024). The Fair Credit Reporting Act extends this to credit-based denials more broadly (Federal Trade Commission 2026). The regulatory intent is straightforward: a customer denied credit is entitled to know why, both to enable challenge of the denial and to inform the customer's understanding of their own creditworthiness. The empty chair the regime addresses is the denied applicant.

In the era of statistical underwriting models — credit scoring, regression-based decisioning, rule-based systems — the adverse action notice was a tractable artifact. The model's decision was based on a small set of factors with weights the institution could inspect; the notice could state the factors that contributed most to the denial and be substantively faithful to the decision. The reasoning was simple enough that the notice's reasons were the model's reasons.

In the era of complex machine-learning underwriting models, the artifact has not changed but the reasoning has. The model evaluates hundreds or thousands of features through nonlinear interactions; the "reasons" extracted for the adverse action notice are post-hoc attributions — typically Shapley values, LIME approximations, or feature importance measures applied locally to the denial decision (Consumer Financial Protection Bureau 2022). The notice states reasons; the reasons are the output of an explanation method, not the reasoning of the model. The empty chair receives an artifact in the form the regulation requires. The artifact does not represent the empty chair's interest in the way the regulation intends.

This is the dishonest middle at its canonical location. The form is satisfied. The reasoning is unverified. The regulatory regime treats the form as evidence of the substance, because the regime was designed for an era when form and substance coincided.

The empty-chair reading of the situation: the denied applicant is named explicitly in the regulation, present in the artifact, and absent from the decision context. Their presence in the artifact is the form of representation; their absence from the decision context is the substance. The frame's diagnostic question — what makes the applicant's interests audible in the architecture — produces an answer the architecture does not currently provide. The applicant receives reasons. The applicant does not receive verifiable reasoning.

The produced-absence dimension, picking up the distinction from Section 2: the applicant who would have challenged the denial if challenge were tractable does not challenge. Adverse action notices are written in regulatory boilerplate, the reasons are typically generic ("credit history," "debt-

to-income ratio," "insufficient income for amount requested"), and the applicant has no mechanism to interrogate whether these reasons accurately describe the model's behavior on their application (Consumer Financial Protection Bureau 2022). The architecture produces silence — applicants who do not appeal because appeal is structurally unavailable — and reads the silence as evidence that the notice was adequate. The institution's position that adverse action notices satisfy ECOA / Reg B is sustained partly by the fact that customers cannot effectively contest the notices' faithfulness, not by the notices being faithful.

3.3 The taxonomic move

The categoricals divide AI applications in lending into two categories, and the empty-chair frame makes the division visible.

Applications where verification is structurally available. Pattern recognition surfacing applications for human investigation. Anomaly detection flagging unusual patterns for underwriter review. Consistency monitoring identifying decisions that deviate from institutional norms. In each, the AI's role is bounded to surfacing; the human investigates the flagged case with full context, reasons through the decision themselves, and produces reasoning the institution can verify because it was the human's reasoning. The empty chair (the applicant) is represented because the human's reasoning is what the adverse action notice describes, and the human's reasoning can be made auditable by capturing it contemporaneously rather than reconstructing it.

Applications where verification is not structurally available. Decision generation by an underwriting model whose output is communicated to the applicant via an adverse action notice extracted by a post-hoc explanation method. The model decides; the explanation method produces text; the human signs. No party has produced reasoning the empty chair can hold the institution accountable for. The institution has produced an artifact and a signature.

These are not gradations along a single axis of risk. They are architecturally different. The first category is AI as decision-support tool with human reasoning as the locus of accountability; the second category is AI as decision-maker with no locus of accountability that survives examination. The line between them is structural — it is determined by where the reasoning lives, not by how risky the decision is.

A common rejoinder: "the human has the authority to override the AI's decision." This is true at the org-chart level and false at the architectural level. Override authority is meaningful only if the human has reasoning of their own to apply against the AI's recommendation. A reviewer presented with a denial recommendation, an explanation, and a signature line does not have time to develop independent reasoning at scale; the architecture has structurally placed the human in a verification role they are not equipped to perform. Override authority is preserved as a procedural fiction. The reasoning is the AI's. The signature is the human's. The accountability locus is nowhere.

3.4 What the categoricals foreclose, and what they do not

The categoricals foreclose architectures that rely on post-hoc explanation as a verification regime in adversarial regulatory contexts. They do not foreclose AI in lending. They foreclose specific architectural patterns:

- Underwriting models whose decisions are communicated via adverse action notices generated by post-hoc explanation methods, with human signatures attesting to review of the notice rather than verification of the reasoning. This is the dishonest middle. The empty chair is unrepresented.

- Decision-recommendation systems where the human's role is signature on AI-generated reasoning rather than independent reasoning the AI surfaces material for. The accountability locus is fictional.
- Documentation regimes that capture human reasoning as post-hoc justification rather than contemporaneous evidence, where the absence of contradicting evidence in the documentation record is treated as evidence the documentation is faithful. The reviewer's actual reasoning is produced-absent.

The categoricals do not foreclose:

- Ante-hoc interpretable models whose reasoning is structurally available for inspection. The model's decision and the model's reasoning are the same artifact; verification is available.
- Decision-support tools where AI surfaces material for human reasoning, the human reasons independently, and the reasoning is captured contemporaneously. The accountability locus is the human; the human's reasoning is verifiable.
- Architectures that bind explanations to underlying reasoning through observable channels — tensor-channel observation, attention pattern attestation, intermediate-representation logging — where the binding can be examined rather than asserted. These architectures are the work of Paper 2; we name the category here as the structural alternative to the dishonest middle.

3.5 A turn: pricing as produced-absence canonical case

The lending example demonstrates the categoricals at their canonical regulatory location, where the dishonest middle takes the form of an artifact (the adverse action notice) that misrepresents the reasoning behind a denial. This is not the only form the dishonest middle takes. We turn briefly to risk-based pricing in consumer lending — the determination of the interest rate, fees, and terms offered to an *approved* applicant — to demonstrate that the frame produces interpretive leverage in regulatory spaces where the empty chair is not addressed by an artifact at all.

The Truth in Lending Act requires disclosure of a rate's mathematical structure (APR, finance charges, payment schedule); it does not require disclosure of why this particular rate was offered to this particular applicant (Consumer Financial Protection Bureau 2026). Unlike the denied applicant, the priced applicant has no regulatory entitlement to a substantive explanation of their rate. The applicant receives a rate; the applicant accepts or declines; the institution proceeds.

The dishonest middle in pricing is more diffuse than in denial because there is no regulatory artifact equivalent to the adverse action notice. The pricing decision is not communicated to the customer as a reasoned determination; it is communicated as a take-it-or-leave-it offer. The customer accepts because they need credit and have no basis for negotiation. The institution reads the customer's acceptance as evidence that the rate was acceptable.

This is produced absence at architectural scale. The customer who would have challenged the rate if challenge were a thing they expected to receive does not challenge, because the architecture of risk-based pricing does not contemplate challenge as a normal part of the transaction. The institution reads the customer's silence — accepted offers, completed transactions, regulatory examinations finding no fair-lending issues — as evidence that pricing is fair. The architecture has produced the silence; the institution interprets the silence as endorsement.

The frame's diagnostic move on this configuration is the produced-absence move from Section 2: *what is the architecture doing that creates this silence, and what is it inferring from the silence it created?*

The architecture is producing pricing decisions through models whose reasoning is not communicated to the customer; structuring the transaction such that the customer cannot effectively negotiate without comparable offers they cannot tractably obtain; and producing aggregate pricing data that institutions analyze for fair-lending compliance, while the data itself is censored — customers who would have asked for justification if justification were available do not exist as a population the institution measures.

The institution then infers from this silence that pricing is fair. The customers who accepted are taken as evidence the rates were acceptable. The customers who declined are taken as having self-selected out. The customers who never applied because they expected unfavorable rates are not measured at all. The architecture has produced a population of accepted-and-silent customers and reads their silence as evidence of pricing fairness.

A common rejoinder: pricing models are statistically validated against outcomes — default rates by pricing tier, fair-lending disparate-impact analysis, ongoing monitoring (Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency 2011; Board of Governors of the Federal Reserve System 2026). This is true and important, and it is not a verification regime for the empty chair we are discussing. Statistical validation against outcomes verifies that the model performs as intended in aggregate; it does not verify that any individual customer's rate reflects their creditworthiness rather than model artifact. The aggregate empty chair (the regulator examining fair-lending compliance) is partly represented; the individual empty chair (the priced applicant) is not. The frame's contribution here is the aggregate-versus-individual distinction the regulatory regime currently elides.

3.6 The taxonomic move, summarized

The lending example demonstrates the categoricals at the regulatory location where the dishonest middle is most directly addressable: where an artifact exists, where the regulation requires the artifact to represent the empty chair, and where post-hoc explanation methods have produced an artifact that does not. The pricing turn demonstrates that the frame extends to regulatory locations where the empty chair has been produced-absent by the regulatory regime itself, and that the frame proposes architectural primitives that would represent the chair the regulation does not currently require representing.

The remaining sections develop the diagnostic techniques that operationalize this taxonomic move (Section 4) and the categories of architectural primitive that empty-chair representation requires (Section 5). The frame's claim is that an honest AI governance framework must hold the structural line, both where the regulation requires it (as in adverse action notices) and where the regulation has not yet recognized that the line exists (as in risk-based pricing). Whether this claim is sustainable in practice is the subject of Section 6's honest uncertainties.

4 Diagnostic Techniques

The empty-chair frame produces interpretive structure (Section 2) and a structural taxonomy of AI applications by verification-regime availability (Section 3). The three positions earned by the body sections — that governance which does not induce auditable constraint is theater, that post-hoc explanation is structurally inadequate as a verification regime in adversarial contexts, and that human review of explanations is not verification of reasoning — appear at first as three separate critiques of three different practices. They are not. They are three sites where the same structural failure mode operates, and naming the underlying mode is what makes the categoricals cohere into a structural argument rather than a list of related concerns. We name the mode *silence-manufacture*

and develop it in this section as the organizing concept beneath the empty-chair frame. The three diagnostic techniques that follow are *silence-breaking instruments* against silence-manufactured artifacts. Their value is not in producing better metrics; their value is in making audible what the architecture has architecturally suppressed.

We frame the techniques from the architect's perspective because the techniques are most powerful when they constrain construction rather than evaluate existing artifacts. An architect who applies them builds systems that are diagnosable; an examiner who applies them to an undiagnosable system can only document that diagnosis is unavailable. The architectural moment is where empty-chair representation is structurally determined, and the diagnostic techniques are most valuable as design-time discipline. The examination-time application of these same techniques falls out as a consequence and is treated below as such.

4.1 Silence-manufacture as the underlying structural pattern

4.1.1 Definition

We define **silence-manufacture** as the structural pattern in which an architecture suppresses access to ground truth in some domain, then produces a substitute artifact — a metric, a document, a signature, an explanation — that occupies the rhetorical position where evidence of ground truth should appear. The artifact is read as evidence; the suppression is read as endorsement; the architecture's continued operation depends on both readings remaining unchallenged.

Silence is the *enabling condition*; substitution is the *active mechanism*. The artifact is real. A Shapley value is a real numerical computation. An audit document is a real institutional production. A signed adverse-action notice is a real legal instrument. None of those facts establish that the artifact binds to what it claims to bind to. The binding is architecturally suppressed. The artifact's apparent epistemic content — *this represents the underlying reasoning, decision-making, oversight* — is sustained by the architecture's suppression of what would falsify the claim.

4.1.2 Relationship to Goodhart's law

Silence-manufacture is the *dual* of Goodhart's law (Goodhart 1984), not strictly its inverse. The relationship is rhetorically useful for orienting readers — Goodhart provides a known landmark — but the load-bearing name is silence-manufacture, not extended-Goodhart.

Goodhart's law describes optimization pressure on a metric corrupting the metric: when a measure becomes a target, it ceases to be a good measure. The pressure lands on the visible; the visible breaks. Silence-manufacture describes suppression pressure on disconfirming evidence sustaining the metric's apparent validity: when silence becomes a measure, it ceases to be evidence of the absence it appears to indicate. The pressure lands on the invisible; the visible appears intact because what would falsify it has been architecturally suppressed.

The two patterns are related through structural symmetry rather than direction-reversal. Goodhart describes corruption of metrics; silence-manufacture describes manufacture of metrics. A regulatory regime that has identified Goodhart's law as the failure mode of its measurement apparatus and tightened its measurements accordingly may still be operating silence-manufacture untouched, because the silence layer is not where Goodhart-shaped reform lands. We return to this implication in the predictions below.

4.1.3 Mapping the three earned positions onto silence-manufacture

Each of the three positions earned by the body sections is a silence-manufacture instance. The mapping is not a rhetorical embellishment; it is the structural claim the section makes.

Position 2: post-hoc explanation as structurally inadequate verification in adversarial contexts. The model's actual reasoning is architecturally inaccessible to the explanation channel. The explanation — Shapley value, LIME approximation, attention-attribution heatmap, chain-of-thought trace — is generated by the same loop that produced the decision, and the binding from explanation to underlying computation is what the architecture suppresses. The explanation occupies the rhetorical position the reasoning should occupy. The artifact is real; the binding is suppressed. The empirical signature is direct: chain-of-thought traces can be perturbed without changing the final answer, and changed without perturbing the final answer (Lanham et al. 2023; Turpin et al. 2023). The explanation is not bound to the underlying computation; it is bound to the appearance of the computation. This is silence-manufacture at the model-output level, with empirical confirmation from a research community — ML interpretability — that has been documenting the phenomenon without naming the structural pattern that links it to compliance theater.

Position 3: human oversight of explanations as not verification of reasoning. The reviewer can only see what the architecture has chosen to expose; the underlying reasoning is suppressed from their view. The signature attests to the human's review of the explanation, not to verification of the reasoning. The signature occupies the rhetorical position of accountability. The empirical signature in ML alignment research is sycophancy and reward-hacking under RLHF: models optimize for signals visible to human evaluators, and human approval becomes a metric for what the model has produced rather than for the model's quality. The architecture trains the model on what the reviewer can see, then reads the reviewer's approval as evidence about what the reviewer cannot see. This is silence-manufacture at the accountability level, with the same structural shape as position 2 — substitute artifact, suppressed binding, the two readings sustaining each other (Sharma et al. 2023; Perez et al. 2023).

Position 1: governance versus governance theater. Compliance theater is the population-level instance of the same pattern. Institutions produce documentation that satisfies a framework; the contemporaneous decision-making the documentation is meant to represent is architecturally suppressed; the documentation is read as evidence of the suppressed decision-making. The empirical signature in adjacent ML literature is benchmark gaming, eval-set contamination, and the Goodhart-on-safety-evals pattern: metrics survive while the underlying construct degrades, and the architecture reads metric-stability as evidence of construct-stability. This is silence-manufacture at the institutional level, structurally identical to its model-output and accountability instances.

The mapping does work the categoricals could not do separately. Three regulatory critiques and three ML interpretability findings turn out to describe the same structural pattern at three different levels of organization. The paper's claim is structural: self-measuring architectures cannot verify themselves; observable failure modes confirm this from both regulatory experience and ML research; two research communities have been describing piecemeal what the empty-chair frame unifies. The cross-domain confirmation strengthens the structural argument substantially, and it makes the unification accountable to evidence from communities that did not produce the frame.

4.1.4 What silence-manufacture predicts

The frame produces predictions that distinguish it from generic Goodhart-shaped critique. We name three.

Goodhart-shaped reform fails predictably. Policy responses that target the visible metric —

tighten the controls, harden the targets, raise the thresholds, sophisticate the scoring — leave silence-manufacture intact. The silence-production architecture manufactures new substitutes that pass the new metrics. Reform succeeds at improving documentation; documentation continues to substitute for ground truth; nothing structural changes. The persistence of compliance theater across reform cycles, in banking and elsewhere, is the prediction operating. The fix has to live at the silence layer — contemporaneous capture, evidence binding, transparency mandates that make the suppressed ground truth inspectable — not at the metric layer. Section 5's architectural primitives are the implementation of this implication; Section 6 marks the prediction's dependence on political-economy assumptions the paper does not establish.

Analytical literatures partition around silence-producing architectures. When an architecture produces silence about its causal mechanism, observers positioned differently relative to the architecture see different absent parties, reconstruct different stories, and the stories do not converge with more research because the architecture continues producing the silence. Each partition is internally coherent; reconstruction requires breaking the silence, not adding observers. Persistent partitioned literatures whose disagreements do not converge with substantial research investment are the signature of operating silence-producing architectures. Section 7 closes on the de-banking discourse as the canonical real-time example of this prediction in operation.

Silence-producing architectures distort political discourse in characteristic ways. When the architecture suppresses the actual cause of harm, narratives that fill the suppression-shaped gap will be wrong in specific ways, and the wrongness will have political consequences the architecture cannot correct. The de-banking discourse exemplifies this prediction: political-debanking framing fills a silence the BSA/AML regime legally produces under 31 USC §5318(g)(2) (United States Congress 2024). The framing is empirically wrong — Anthony (2026) reports that 35 of 8,361 account-closure complaints mention political or religious motivations — but wrong in the shape the silence makes available. The prediction is structural: any architecture that legally or operationally suppresses the actual cause of a politically salient harm should be expected to produce mis-attributing public discourse, and the discourse's mis-attribution is itself diagnostic information about the operating silence.

The three predictions are not independent claims; they are three consequences of one structural pattern, applied to three sites — regulatory reform, analytical inquiry, and political discourse. The empty-chair frame's load-bearing structural commitment is silence-manufacture; the predictions are how the frame earns the unification rather than merely asserting it.

4.2 Three diagnostic techniques

The diagnostic techniques developed below are silence-breaking instruments. Each technique is paired to a class of silence the architecture produces, and the technique's value is in what it makes audible. We claim three techniques because they emerge cleanly from the frame's structural commitments and produce design constraints directly applicable to AI governance system construction. Other techniques are possible and likely; these three are the ones the frame produces first, and together they cover the silence-types the three earned positions identify.

4.2.1 Technique 1: Empty-chair enumeration as design pre-condition

Before specifying any architectural primitive, the architect asks: *which empty chairs is this system supposed to represent, and what does representation require for each?* The question precedes any technical decision. An architecture whose empty-chair attribution is left implicit will produce primitives that serve some chairs and fail to serve others, and the failure will not be visible in

the architecture's documentation because the documentation will describe what the system does, not what it leaves out. The silence the technique breaks is the silence around *which absences the architecture is making consequential*: an architecture that has not enumerated its empty chairs is suppressing the question of which chairs it serves, not answering it.

Section 2 demonstrated this technique applied analytically to BSA/AML transaction monitoring. The same move applies to design. The architect designing an AI governance system for a community bank's lending operation asks first: *whose interest is this system structurally protecting?* The answer is rarely on the page initially; it has to be derived from the regulatory regime, the institution's risk profile, and the categories of decision the system will affect. Making the derivation explicit at design time is the technique's primary work.

The technique produces three outputs at design time:

First, an enumeration of empty chairs the system affects. This is the analytical move from Section 2, applied at the moment of architectural commitment.

Second, a mapping from empty chairs to architectural requirements that representation imposes. Different empty chairs require different architectural support. The future compliance officer needs durable, queryable institutional knowledge. The denied applicant needs verifiable reasoning. The marginalized customer needs disparate-impact monitoring with population decomposition. The institution's future self needs tamper-evident temporal capture. Each chair generates a set of requirements; the requirements may conflict, and the conflicts are themselves design information.

Third, a constraint on architectural choices: any primitive that fails to satisfy at least one chair's requirement and creates costs without serving any chair is architectural overhead. Any primitive that satisfies one chair's requirement at the cost of failing another's is a design tradeoff that requires explicit attention. The architecture is not a set of capabilities; it is a set of empty-chair representations, with capabilities chosen as the means.

The technique's design effect is to shift architecture decisions from feature-driven to representation-driven. Architectures designed without the technique tend to accumulate capabilities — explainability features, audit trails, monitoring dashboards — without explicit attribution to which empty chairs the capabilities serve. Architectures designed with the technique start from the chairs and select capabilities as the means of representing them. The selection criterion makes capability decisions tractable; without the criterion, capability decisions reduce to keeping up with vendor offerings.

4.2.2 Technique 2: Designing for pages missing from the ledger

A well-designed attestation regime makes the institution's reasoning inspectable. But attestation cannot be uniformly comprehensive; cost and operational reality require choices about what to attest and what to leave un-attested. The architect's diagnostic move at design time is to make these choices explicitly and to design the attestation regime such that *what is left out is itself informative*. The silence the technique breaks is the silence between the institution's posture and its documentary record: under ambient absences, posture is unreadable; under documented absences, posture becomes a pattern an examiner can read.

We call this technique *designing for pages missing from the ledger*. The metaphor is deliberate: an accounting ledger with missing pages is more diagnostic than a forged ledger if the pattern of missing pages is itself interpretable. A ledger where missing pages are random reveals nothing about the institution's posture; a ledger where missing pages cluster around decisions favorable to the institution and unfavorable to challenged customers reveals the institution's posture clearly.

The technique works as follows. The architect determines, for each category of decision the system handles, what attestation will and will not be produced. The determination is documented. The institution's choice not to attest a category is itself attested, with the rationale captured. An

examiner reading the system's attestation later can read both the present attestation (what was captured) and the meta-attestation (what was deliberately not captured, and why). The pattern of present and absent attestation is interpretable because the absences are themselves documented choices rather than oversights.

This is a design discipline, not a construction discipline. Most attestation systems are built bottom-up: a capability is implemented, attestation for that capability is added, and the absence of attestation in other areas is ambient. Designing for pages-missing inverts this: the architect specifies in advance the full domain of decisions, marks which will and will not be attested, and ensures the system's structure makes the absence visible rather than ambient.

The technique produces a design output: a manifest of attested and un-attested decision categories, with rationale, that travels with the system as part of its architectural documentation. An examiner reading the manifest can ask whether the un-attested categories are operationally explicable (resource constraints, low-risk decisions deprioritized) or whether they cluster in ways that reveal a posture (decisions favorable to the institution attested, decisions unfavorable to challenged customers not). The architect who has produced the manifest knows in advance which questions the manifest will answer and which it will not, and can adjust the attestation scope accordingly.

The architect's question to themselves at this stage: *what does my pattern of attestation reveal about my posture toward empty-chair representation, and is that what I intend it to reveal?*

4.2.3 Technique 3: The produced-silence question pair as design constraint

Section 2 introduced the distinction between structural and produced absence. The diagnostic technique on produced absence, applied at design time, is a question pair the architect asks of every interface, every documentation regime, every customer-facing artifact:

What is this design doing that creates silence?

What is this design inferring from the silence it creates?

This is the silence-manufacture-direct technique. The first two techniques operate on silences that the architecture has implicitly allowed to form (empty chairs not enumerated, decision categories ambient about their attestation status); this technique operates on silences the architecture *actively produces*, then *reads as endorsement*. The two-part question maps directly onto the two parts of silence-manufacture: the first question surfaces the architecture's silence-production, the second surfaces the substitute artifact and the inference being drawn from suppression.

A design that asks reviewers to write justification after deciding creates silence around the reviewer's contemporaneous reasoning, then infers from the absence of contradicting evidence in the post-hoc justification that the documentation is faithful. The architect applying the question pair at design time recognizes the silence-production and chooses differently — capturing decision context as the reviewer engages with it rather than asking for narrative reconstruction.

A design that produces adverse action notices in regulatory boilerplate creates silence around the customer's effective interrogation of the notice, then infers from the absence of customer challenges that the notice is adequate. The architect applying the question pair at design time recognizes the boilerplate's interrogation-resistance and chooses differently — producing notices the customer can effectively interrogate, even if doing so requires architectural work the regulation does not currently require.

A design that prices loans without communicating substantive justification to the customer creates silence around the customer's standing to negotiate, then infers from acceptance rates that pricing is fair. The architect applying the question pair at design time recognizes the absence of customer-side justification mechanisms as a design choice with consequences, and weighs whether the consequences are acceptable given the empty chair the design leaves unrepresented.

The question pair, applied at design time, has the same structure as at examination time but different effect. At examination time, it diagnoses what the architecture has already done. At design time, it constrains what the architecture will do. The architect who anticipates the diagnostic question pair builds a system that can answer it; the architect who does not builds a system that the question pair will diagnose negatively.

This technique is the most contestable of the three because it requires the architect to design for absences they are choosing to allow. An architect may legitimately respond that some silences are structural rather than produced, that customers who don't challenge are satisfied, that reviewers who don't surface contemporaneous reasoning don't have any to surface. The architect may be correct in some cases. The technique's value is in making the question explicit at design time so the architect's choice is conscious rather than ambient.

The architect's question to themselves at this stage: *for every silence my system produces, am I confident the silence reflects the absence of objection rather than the suppression of objection by the system itself?*

4.3 The techniques as compounding design constraints

The three techniques compound at design time. Empty-chair enumeration breaks the silence around which absences the system is structurally responsible to. Designing-for-pages-missing breaks the silence between institutional posture and documentary record. The produced-silence question pair breaks the silence the architecture itself manufactures and reads as endorsement. In combination, the techniques do not merely produce a checklist of design requirements; they produce a design discipline whose effect is to replace the architecture's manufactured silences with inspectable evidence at the points where silence-manufacture would otherwise sustain an unchallenged substitute.

This is structurally different from feature-driven architecture. The system's capabilities are chosen as means of empty-chair representation; the system's attestation regime is designed for pattern-readability; the system's interfaces are scrutinized for silence-production at the moment of design rather than at the moment of examination. The architecture that results is one that an examiner can read; the architecture without these techniques is one that an examiner cannot read regardless of skill.

This is the operational claim of the section: the empty-chair frame, developed into design-time diagnostic techniques against silence-manufactured artifacts, produces architectural discipline that makes the resulting systems substantively examinable. The frame is not an examination methodology that compensates for inadequate architecture; it is an architectural discipline that produces examinable systems. The examination-time application of the techniques follows from the architectural use, not the other way around.

4.4 Examination as consequence

The same three techniques, applied by an examiner to an existing system, produce assessment leverage the FS AI RMF's checklist does not provide. The examiner asks: *which empty chairs does this system claim to represent? What does its attestation pattern reveal about its actual representation? What silences does its architecture produce, and what does the architecture infer from those silences?* These are the same questions the architect asked at design time, redirected from "what should I build" to "what was built."

The asymmetry between design-time and examination-time application is important. A system designed with the techniques in mind answers all three questions; an examiner reading such a system can assess whether the answers are substantive or theatrical. A system designed without the

techniques in mind may be unable to answer the questions at all; an examiner reading such a system can only document the unavailability of substantive assessment.

The implication: the techniques' regulatory effect depends on adoption at the architectural level, not just at the examination level. An examiner who applies the techniques to a system designed without them produces findings of inadequacy, but the inadequacy is structural — it cannot be remediated by additional examination, only by re-architecture. The frame's claim is that AI governance systems for community banking should be architected to be diagnosable, and the diagnostic techniques developed here should shape the architecture rather than only its evaluation. The development of examination-time methodology is the work of Paper 3; this paper's diagnostic register is architectural because the architectural moment is where the techniques have most leverage.

4.5 Implications for architectural primitive selection

Section 5 develops the categories of architectural primitive that empty-chair representation requires. The diagnostic techniques developed here constrain that development: a primitive is valuable in proportion to its support of design-time diagnostic discipline against silence-manufacture. Tamper-evident temporal capture is valuable because designing-for-pages-missing requires the ability to attest both presence and absence of records, and because the silence-manufacture pattern at the institutional level is sustained by retroactive narrative substitution that contemporaneous capture forecloses. Three-dimensional confidence characterization is valuable because empty-chair enumeration requires distinguishing decisions where the architect provided genuine evidentiary support from decisions where the architect produced narrative without substantive grounding — the distinction silence-manufacture exists to obscure. The primitive categories are not generic technical capabilities; they are capabilities selected because the design-time techniques require them, and the techniques require them because silence-manufacture is the failure mode they are constructed against.

This is the section's transition into Section 5. The techniques operationalize the frame at design time; the primitives operationalize the techniques. Each layer of the architecture is justified by what the layer above it requires of it, and the layer above the techniques is the structural pattern they are constructed to break.

5 Architectural Primitive Categories

The empty-chair frame produces interpretive structure (Section 2), a structural taxonomy of AI applications (Section 3), and diagnostic techniques that operationalize the frame as design discipline and examination methodology (Section 4). This section names the architectural primitive categories that empty-chair representation requires. The treatment is at the property level: each primitive is described by what it must do, not by how it must be implemented. Implementation specification is the work of follow-on papers, and we mark places where the discussion is at risk of crossing into implementation territory so that the property-level claims remain the substance of this section.

A framing note before the primitives are developed. The four primitives below are *silence-breaking instruments*, not improvements to existing metrics. Read through a Goodhart lens — "the controls became targets, so we need better controls" — they appear underwhelming, because Goodhart-shaped readings expect the fix to be at the metric layer (Goodhart 1984). The frame's diagnosis (Section 4) is that the failure mode is silence-manufacture: the architecture suppresses access to ground truth and produces a substitute artifact that occupies the rhetorical position evidence should occupy. The fix has to live at the silence layer — making the suppressed ground truth contemporaneously inspectable, traversable, and characterized honestly. Each primitive corresponds to a specific class of silence the architecture currently produces, and the primitive's value is in breaking it. We mark

this here because readers approaching the section without the frame's diagnostic structure are likely to mis-read the primitives as competing implementations of better controls, and the mis-reading produces frustrated assessments of adequacy that the structural argument does not warrant.

The primitives are not exhaustive. They are the categories the diagnostic techniques in Section 4 require. An honest AI governance system for community banking will likely require additional primitives we have not identified; the four named here are the categories we can argue for from the frame's structural commitments alone.

5.1 Primitive 1: Tamper-evident temporal capture

The property: the architecture captures decision context at the moment the decision is made, in a form that binds the captured context to the moment of capture, that prevents retroactive alteration of either the context or the temporal binding, and that can be later inspected without reliance on the institution's reconstruction of what was captured.

The diagnostic technique that requires this primitive: pages missing from the ledger (Section 4, Technique 2). The technique reads patterns of presence and absence in attested decisions as substantive information about the institution's posture. The reading requires that present attestations be trustworthy — that an attestation present in the record was actually produced at the time it claims, contains what it claims to contain, and has not been altered since. Without these guarantees, present attestations are no more diagnostic than absent ones; the institution can produce attestations after the fact that resemble contemporaneous capture.

The property has three components. First, *contemporaneous capture*: the attestation is produced as the decision is made, not reconstructed afterward. Second, *temporal binding*: the attestation's claim to a particular time of capture is itself attested in a way that resists later alteration. Third, *content binding*: what the attestation says was captured cannot be changed after capture without the change being detectable.

The empty chair this primitive most directly serves is the institution's future self. An examiner arriving in three years to review decisions made now can rely on contemporaneous attestation in a way they cannot rely on reconstructed narrative; the future institution's exposure to enforcement risk from gradually-degraded documentation is structurally addressed by capture that cannot be retroactively improved.

The primitive also serves the future compliance officer (institutional knowledge survives personnel turnover in a form that does not depend on the departing officer's narrative reconstruction) and indirectly serves the customer affected by an AI decision (the customer's challenge to a decision made years prior can rely on what the institution attested at the time of decision, rather than on what the institution claims now to remember).

[QUARRY: PAPER 2] Implementation choices for tamper-evident temporal capture include cryptographic anchoring to public timestamp services, append-only event-sourced storage with cryptographic chaining, distributed attestation across nodes with byzantine fault tolerance, and various combinations of these. Different implementations have different cost profiles, different examination-time inspection mechanisms, and different exposure to specific attack classes. Paper 2 specifies the implementation tradeoffs at the level of architectural commitment.

The property-level claim is that any honest implementation must satisfy contemporaneous capture, temporal binding, and content binding. Implementations that satisfy two of three are not partial successes; they are failures in the dimension they fail. An attestation regime with content binding and temporal binding but without contemporaneous capture allows reconstructed narrative to be cryptographically signed; the signature attests to the institution's commitment to the narrative, not to the contemporaneous accuracy of the captured decision context. This is a sophisticated form

of the dishonest middle that the primitive's three-component definition is meant to foreclose.

5.2 Primitive 2: Evidence binding

The property: the architecture binds decisions to the evidence that supported them, in a way that the binding can be later inspected, traversed, and verified. A decision attested without evidence is not the decision the empty chair needs to inspect; the empty chair needs the decision and the chain of evidence the decision rested on.

The diagnostic technique that requires this primitive: empty-chair enumeration (Section 4, Technique 1). The technique requires that, for each empty chair a control was meant to protect, the institution can demonstrate evidence that the implementation represents that chair. Demonstration requires evidence; evidence requires that the architecture preserved the chain from decision back to supporting material. Without evidence binding, demonstration reduces to assertion.

The property has two components. First, *evidence preservation*: the architecture preserves the material that supported each decision in a form that survives institutional turnover, system migration, and ordinary records management. Second, *traversal capability*: an examiner or other inspector can walk from a decision backward through the evidence that supported it, and forward from a piece of evidence to the decisions that relied on it. The two-direction traversal matters because empty-chair questions arrive in both forms — *what evidence supported this decision* and *what decisions depended on this evidence source*.

The empty chair this primitive most directly serves varies by which decision is being inspected. For an adverse action notice, the empty chair is the denied applicant, and evidence binding allows inspection of what the institution actually relied on (which may or may not match the post-hoc explanation produced for the notice). For a loan-pricing decision, the empty chair is the priced applicant, and evidence binding allows inspection of which underwriting features actually drove the rate. For a SAR filing decision, the empty chair is shared between the next victim of fraud (whose protection depended on the filing) and the customer who was reported (whose interest is in whether the filing was substantively warranted).

[QUARRY: PAPER 2] Implementation choices for evidence binding include graph-structured evidence stores with explicit edges between decisions and supporting material, content-addressed storage where evidence is referenced by cryptographic hash of content, lineage-tracking systems that maintain provenance metadata across transformations, and various hybrid approaches. The implementation choice has substantial implications for query performance, storage cost, and the granularity at which evidence can be traversed. Paper 2 specifies these tradeoffs.

The property-level claim is distinct from logging: a system that logs decisions and logs evidence but does not bind them is not satisfying this primitive. The binding is what enables traversal. A pile of documents and a pile of decisions, both well-attested, are not evidence-bound unless the architecture preserves the relation between them.

A specific structural concern: evidence binding interacts with post-hoc explanation methods in a way the primitive must address explicitly. A system that binds the post-hoc explanation to the decision satisfies the primitive's literal requirement while undermining its substantive purpose. The empty chair inspecting the bound evidence finds a Shapley-value attribution; the attribution was bound to the decision; the binding is verifiable. The primitive is satisfied. The empty chair has not been served. The honest formulation of the primitive requires that the bound evidence be the *underlying reasoning* the decision was based on, not a post-hoc reconstruction. This is why the primitive cannot be specified in isolation from the categoricals from the opening: the primitive's property-level definition must include the categoricals, or the primitive can be implemented in ways that satisfy its form without serving its function.

5.3 Primitive 3: Three-dimensional confidence characterization

The property: the architecture characterizes the epistemic state of every assessment along three dimensions — confidence (evidence supports the conclusion), concern (evidence supports an alternative conclusion), and indeterminacy (evidence is insufficient to determine either) — rather than collapsing epistemic state to a single confidence score.

The diagnostic technique that requires this primitive: empty-chair enumeration applied to representation evidence (Section 4, Technique 1). The technique asks whether the institution has evidence that the implementation represents each empty chair. The honest answer to this question, for many cases, is "we do not know" — the institution's data does not establish that representation occurred, but neither does it establish that representation failed. The honest answer is indeterminacy, not confidence in a particular direction. A confidence-only architecture cannot express indeterminacy; it must report a confidence value, and the confidence value is then read as the institution's belief about the case, when the institution's actual epistemic state was ignorance.

The property has three components corresponding to the three dimensions. *Confidence* measures evidence supporting the conclusion. *Concern* measures evidence supporting alternatives. *Indeterminacy* measures the insufficiency of evidence to determine either. The three are orthogonal: high confidence and high concern simultaneously indicates a contested case where evidence pulls both ways; low confidence and low concern with high indeterminacy indicates a case the institution does not have grounds to evaluate.

The empty chair this primitive serves is the affected party who depends on the institution's epistemic state being honestly characterized. A customer denied credit on a high-confidence assessment can challenge the assessment substantively. A customer denied credit on a high-indeterminacy assessment that was reported as moderate confidence has been deceived about the institution's actual state; the institution's reporting collapsed indeterminacy to a confidence value the institution did not actually have grounds for.

[QUARRY: PAPER 2] Implementation choices for three-dimensional confidence characterization include neutrosophic-set-theoretic representations, evidence-theoretic frameworks (Dempster-Shafer and related), explicit indeterminacy flags layered on conventional confidence scores, and various other approaches. The choice has implications for how the three dimensions are computed, how they are reported in artifacts the empty chair encounters, and how they propagate through chained assessments. Paper 2 specifies these.

The property-level claim is that any honest implementation must distinguish all three dimensions; an implementation that maps everything to a single confidence value is not satisfying this primitive even if the value is computed sophisticatedly. The distinction matters most in the cases where institutions are most tempted to elide it: cases where the institution's actual epistemic state is indeterminacy but where reporting indeterminacy is operationally inconvenient. Those are precisely the cases where the primitive's diagnostic value is highest.

5.4 Primitive 4: Structural drift typology

The property: the architecture monitors and reports drift along distinct types — data drift, concept drift, performance drift, and institutional drift — rather than collapsing drift to a single performance metric.

The diagnostic technique that requires this primitive: produced-silence question pair (Section 4, Technique 3). The technique asks what silences the architecture is producing and what it is inferring from those silences. Drift typology is structurally relevant because different drift types reveal different architectural patterns of silence-production. Performance drift may indicate model degradation; data

drift may indicate population shifts the model was not trained for; concept drift may indicate that the relationships the model was trained on have changed; institutional drift may indicate divergence between de facto practice and de jure policy. Each is a different architectural fact about what the institution's deployment is doing, and each has different implications for empty-chair representation.

The property has four components corresponding to the four drift types. *Data drift* tracks distributional changes in inputs. *Concept drift* tracks changes in the relationship between inputs and the phenomena they predict. *Performance drift* tracks changes in the model's outputs against ground truth or proxies. *Institutional drift* tracks divergence between the institution's documented policy and its operational practice as captured in the attestation regime.

The empty chair varies by drift type. Data drift most directly affects the marginalized customer (whose population may have shifted in ways the model does not track). Concept drift affects the institution's future self (whose enforcement exposure depends on whether the model's predictions remain valid). Performance drift affects the next victim of fraud or the false-positive customer (whose exposure depends on the model's continued accuracy). Institutional drift affects the future compliance officer (who needs to understand how practice diverged from policy) and the examiner (whose review requires that documented policy match operational practice).

[QUARRY: PAPER 2] Implementation choices for drift detection include statistical hypothesis testing on input distributions, embedding-space change detection, performance comparison against rolling baselines, and various meta-monitoring approaches that detect drift in the drift detection itself. Paper 2 specifies these.

The property-level claim is that an architecture monitoring "drift" without distinguishing the four types is monitoring a composite signal whose components require different responses. A performance-drift response (recalibration, retraining) is not a data-drift response (population analysis, training-data update) is not an institutional-drift response (policy review, attestation-pattern audit). Conflating them is the architectural equivalent of treating a fever as the underlying disease.

5.5 What this section does not include

The strawman material that informs this section includes a fifth category — institutional self-knowledge — that we have not promoted to a primitive in this section. Our reading is that institutional self-knowledge is not a primitive in the same sense as the four above; it is an *emergent property* of the four primitives operating together over time. An institution with tamper-evident temporal capture, evidence binding, three-dimensional confidence characterization, and structural drift typology accumulates institutional self-knowledge as a byproduct of the four primitives' operation. Naming it as a fifth primitive risks confusing the goal with the mechanism.

This is a structural call we make explicit because a reader familiar with the strawman may expect the fifth primitive and notice its absence. The choice is open to revision; if the four primitives produce institutional self-knowledge less robustly than this section claims, a fifth primitive may need to be specified. Paper 2's implementation work will surface whether the byproduct emerges as claimed.

We have also deliberately not included primitives that would specify the AI components themselves — model architecture choices, training methodology, ante-hoc interpretability techniques, tensor-channel observation mechanisms. By *tensor-channel observation* we mean direct observation of model internal state — activations, attention patterns, intermediate representations — as distinct from text-channel verification, which works only on the model's generated text outputs and inherits the unfaithfulness problems of post-hoc explanation discussed in Sections 1 and 4. These internal-state mechanisms are valuable and the paper's argument leans on some of them (particularly tensor-channel observation as the structural alternative to text-channel verification). But specifying them

as primitives in this section would commit the position paper to claims about what AI systems should look like internally, which is a different argument than the architectural-governance argument the paper is making. The AI components are the subject of separate research; the governance architecture is what this paper specifies.

5.6 What the four primitives compound to

The four primitives, operating together, produce an architecture that supports the diagnostic techniques of Section 4 and that serves the empty chairs enumerated across Sections 2 and 3. Tamper-evident temporal capture makes the institution's reasoning inspectable; evidence binding makes the inspection traversable; three-dimensional confidence characterization makes the inspection's findings honest about institutional epistemic state; structural drift typology makes the inspection's findings useful across time as conditions change.

This is the architectural commitment empty-chair representation requires. The commitment is at the property level; multiple coherent implementations of each primitive exist; the choice among implementations is the work of follow-on papers and of the architects who will build community banking AI governance systems under FS AI RMF compliance pressure.

The primitives' substantive contribution is structural rather than methodological. An architecture without these primitives cannot represent the empty chairs the FS AI RMF was written to protect, regardless of how thoroughly the architecture documents its compliance. An architecture with these primitives can represent the empty chairs, even where current regulatory text does not require the representation to be substantive. The primitives are the structural minimum for honest empty-chair representation in AI governance; they are not the maximum, and Paper 2's implementation work will identify additional primitives the implementation reveals as necessary.

The honest uncertainty for this section, picked up in Section 6: we have argued for the four primitives from the structural commitments of the empty-chair frame and the diagnostic techniques the frame produces. We have not demonstrated that an implementation of the four primitives is operationally tractable for a community bank under cost and resource constraints. The frame's claim is structural; the implementation's tractability is empirical, and the empirical question is the work of Paper 2 and of subsequent deployment validation. The primitives' value in this paper is as the structural argument; their viability in practice is a question we do not resolve here.

6 Honest Uncertainties

A position paper that claims certainty everywhere is not to be trusted. The frame, the categoricals, the diagnostic techniques, and the architectural primitives developed in this paper are advanced as defensible structural commitments, not as solved problems. This section names what the paper has not resolved, what its structural arguments do not establish, and where the load-bearing claims rest on conditions the paper cannot verify. The point is not to weaken the paper's positions but to mark where the positions stop, so that subsequent work — whether by these authors or others — can identify the next questions accurately.

We organize the uncertainties by the level of the paper's argument they affect: the frame itself, the categoricals, the diagnostic techniques, and the architectural primitives. A final uncertainty concerns the paper's overall theory of how a position paper produces effect, which is the meta-level concern that the substantive uncertainties fold into.

6.1 Uncertainties about the frame

The empty-chair count is open. The frame names absent parties as the organizing principle but does not specify a closed enumeration of which parties count as empty chairs in any given regulatory context. Section 2's six chairs in BSA/AML monitoring are illustrative, not exhaustive. Different analysts using the frame may identify different chairs, weight them differently, and reach different conclusions about whether a given control's implementation represents the chairs adequately. The frame does not provide a procedure for determining which chairs are present in a given context; it provides a vocabulary and a set of diagnostic questions that, applied conscientiously, surface chairs that would otherwise be implicit.

This is not a fatal indeterminacy. Most useful analytical frames are open in this way; the alternative is a frame so specified that it cannot be applied to novel situations. But the openness is real, and it has consequences. Two examiners using the frame on the same institution may disagree about which chairs the institution is failing to represent. The disagreement may be productive — surfacing different aspects of the institution's posture — or unproductive, depending on whether the examiners can articulate the chairs they are weighting and the reasoning behind the weighting. The frame's value depends on practitioners developing the discipline to make their chair-attribution explicit, which is a craft skill the paper does not specify and that practice will have to develop.

The structural-versus-produced absence distinction may not be exhaustive. Section 2 introduced the distinction; subsequent sections relied on it. The distinction is sharp where it applies, but we have not argued that all absence falls into one of the two categories. Hybrid cases — partly structural, partly produced — appear in Section 2's enumeration, and the paper handles them by marking them as mixed. Whether all absence can be analyzed in terms of these two categories, or whether further categories (collective absence, structural-but-architecturally-correctable absence, others) are needed, is an open question the paper does not resolve.

The frame's generalization beyond community banking is asserted but not demonstrated. We have argued that the empty-chair frame applies to AI governance in regulated decision contexts generally, with examples drawn from community banking. We have not demonstrated the frame's application to insurance underwriting, to healthcare AI, to criminal justice algorithms, to social-services determinations. The frame's structural commitments suggest it should generalize, but the generalization is a hypothesis the paper does not test. Other regulatory contexts may have empty chairs the banking frame does not anticipate; the diagnostic techniques may need adaptation; the architectural primitives may need supplementation. The paper's claims about generalizability are best read as invitations for sectoral application rather than as established results.

Silence-manufacture as the underlying structural pattern increases the paper's coherence and its brittleness simultaneously. Section 4 names silence-manufacture as the structural pattern unifying the three earned positions: an architecture suppresses access to ground truth, then produces a substitute artifact that occupies the rhetorical position where evidence of ground truth should appear, with the artifact read as evidence and the suppression as endorsement. Naming this pattern is what makes the three positions cohere into a structural argument rather than a list of related concerns; it is also what makes the paper vulnerable to single-point rejection. A reader who accepts the empty-chair frame but rejects silence-manufacture as the underlying pattern can still accept the individual categoricals on their own terms, but the paper's full structural force does not land. We have judged the trade-off worth making — the alternative is being read as a list, which is worse for a position paper than being brittle in one specific way — but the brittleness is real and worth marking.

Silence-manufacture is unusually co-optation-resistant, but only at the structural level. The unification is subject to the standard vocabulary co-optation risk: any term load-bearing

enough to organize a structural claim is load-bearing enough to be deployed by the institutions whose practices it critiques. The standard pattern is for the critical term to be attached to a documentary or procedural artifact the institution produces in response (an audit report, a certification, a compliance line item), with the artifact occupying the rhetorical position the original critique opened. The term's critical edge migrates onto the substitute and dulls.

Silence-manufacture is positioned differently. The pattern the term names *is* the production of substitute artifacts that occupy positions where evidence of architectural change should appear. A documentary response to a silence-manufacture diagnosis (an attestation that an institution's controls do not exhibit the pattern, a checklist item under a future framework) is, by the term's own definition, a substitute artifact occupying the position where structural change should be, with the suppression of architectural change read as endorsement. The frame diagnoses its own co-optation. The only response that does not enact the pattern is architectural: actually breaking the silence-producing architecture so the suppressed ground truth becomes inspectable.

This is a property of the term's structural commitment, not a guarantee of practical resistance. Institutions can adopt the vocabulary without adopting the discipline; the term can drift into use as a synonym for "compliance failure" or "documentation gap" with the structural claim discarded. What the property establishes is narrower: such usage is detectable on the term's own grounds, because silence-manufacture rhetoric without architectural substance is itself the pattern the term names. The brittleness trade-off the unification accepts therefore differs from the simpler trade-off described above. The unification is brittle to a reader who rejects silence-manufacture as a structural pattern and reads the categoricals as a list. It is unusually resistant, at the structural level if not at the level of practical adoption, to the more common failure mode of critical terms, which is capture through substitution.

6.2 Uncertainties about the categoricals

The categoricals are structural arguments, not empirical claims. Position 2 (post-hoc explanation as inadequate verification regime in adversarial contexts) and Position 3 (human oversight of explanations as not verification of reasoning) are argued from the structural relationship between explanation methods and model reasoning, with citations to empirical work that supports the argument. The categoricals are not themselves empirical claims; they are structural claims that empirical work has corroborated in specific instances. Whether the categoricals hold in their fully general form — across all explanation methods, all model architectures, all adversarial contexts — is a stronger claim than the empirical work has established. We argue for the structural form because the structural argument is what the paper's other claims rest on; we acknowledge that the structural form has not been demonstrated empirically across all the configurations it claims to apply to.

The "structurally adversarial" characterization of bank examination is a claim about institutional dynamics, not a definitional truth. Examination practice involves adversarial elements (the examiner is testing institutional claims) and cooperative elements (the examiner and institution share an interest in the institution's continued operation under regulatory authority). The paper characterizes examination as structurally adversarial because the verification-regime question requires the adversarial framing to land, but a more nuanced reading would distinguish adversarial examination contexts (where the institution and examiner have genuinely opposed interests on a specific question) from cooperative examination contexts (where they are jointly working out how to address a regulatory question). The categoricals as stated apply to adversarial examination contexts; whether they apply with the same force to cooperative contexts is less clear. We have not developed this distinction in the paper because it complicates the categoricals without changing the load-bearing claims; we mark it here because subsequent work may need to address it.

The categoricals' regulatory uptake is a prediction the paper makes implicitly. The conclusion claims that frameworks built on the dishonest middle "will not survive the recognition" of regulatory scrutiny that learns to ask what an honest verification regime requires. This is a prediction about regulatory direction. The prediction is consistent with the trajectory of the FS AI RMF and with the April 2026 revised model-risk guidance, but it remains a prediction (Sullivan & Cromwell LLP 2026). If regulatory direction shifts — if explainability-as-Shapley-attribution becomes regulatorily entrenched, if "human in the loop" becomes a sufficient condition for AI governance approval, if the GenAI exclusion in current model-risk guidance becomes permanent rather than temporary — the categoricals may remain structurally true while becoming regulatorily orphaned. The paper's political-economy assumption is that the regulatory community is moving in the direction the categoricals point. If that movement stalls or reverses, the paper's regulatory effect may be substantially smaller than the conclusion suggests.

The frame predicts the failure mode of Goodhart-shaped regulatory reform; the prediction depends on political-economy assumptions the paper does not establish. (Goodhart 1984) The silence-manufacture frame predicts that policy responses targeting the visible metric layer — better controls, harder-to-game targets, tightened thresholds, more sophisticated scoring — will leave the silence-production architecture intact, which manufactures new substitutes that pass the new metrics. Reform succeeds at improving documentation; documentation continues to substitute for ground truth; nothing structural changes. The persistence of compliance theater across reform cycles, in banking and elsewhere, is consistent with this prediction. The prediction is not, however, an established empirical result: it depends on assumptions about how regulatory communities respond to silence-producing architectures, on whether silence-breaking interventions (transparency mandates, contemporaneous capture requirements, evidence binding) are politically tractable in the contexts where they would have effect, and on whether the regulatory apparatus is structurally capable of intervention at the silence layer rather than the metric layer. The paper's working assumption is that silence-layer intervention is becoming politically and technically tractable, and that the FS AI RMF's emphasis on architectural compliance over documentary compliance is evidence in favor of that assumption. If the assumption is wrong — if regulatory political economy reliably routes reforms toward metric-layer fixes regardless of structural diagnosis — the frame's predictive content remains correct as a structural claim but produces no policy purchase. The paper does not establish the political-economy assumption; it relies on it. Subsequent work, particularly the examination-methodology paper distinguished as Paper 3, will need to address the tractability question more directly than this paper has.

6.3 Uncertainties about the diagnostic techniques

The techniques' adoption depends on examination resources we have not analyzed. Section 4 names the coordination problem that the techniques' value depends on examination resources, expertise, and institutional support. The paper does not analyze whether these resources are likely to be available. Examination practice is resource-constrained; not every examiner can apply techniques of this depth to every institution. The techniques work best when their existence is widely known across the regulatory community, so that institutions anticipate them and architect for them in advance, rather than relying on examiner application case by case. Whether the techniques achieve sufficient regulatory currency to produce architectural-anticipation effects, or whether they remain niche analytical tools applied by some examiners but not others, is a coordination problem the paper does not address.

The institutional-evasion problem is real and not solved. Sophisticated institutions can architect their systems and documentation to anticipate the diagnostic techniques while preserving the

substantive gaps the techniques are meant to surface. An institution that understands pages-missing diagnosis can produce attestation patterns that appear non-suspicious without actually attesting the substantive material; an institution that understands produced-silence diagnosis can produce challenge channels that appear available without actually being usable. The defense against evasion is examiner expertise developed over time, examination practice that compounds across institutions, and continued evolution of the techniques as institutional responses develop. None of these defenses is automatic, and the techniques' effectiveness in the medium term depends on whether the regulatory community develops them as a community rather than as individual examiners working in isolation.

The aggregate-versus-individual representation argument from the pricing turn is philosophically sharp and not fully developed. Section 3's pricing turn argues that aggregate statistical validation does not constitute verification regime for the individual empty chair. The argument is correct as far as it goes, but it depends on accepting that individual representation has a different weight than aggregate representation, which is not obviously a position the FS AI RMF takes. We have not developed the philosophical argument that grounds the aggregate-versus-individual distinction; we have used the distinction to make the pricing-turn argument land. Whether the regulatory community accepts the distinction as substantively important is uncertain.

6.4 Uncertainties about the architectural primitives

The four primitives may be insufficient. Section 5 names tamper-evident temporal capture, evidence binding, three-dimensional confidence characterization, and structural drift typology as the primitives the diagnostic techniques require. We have argued that these four are necessary; we have not argued that they are sufficient. Implementation work in Paper 2 will reveal whether additional primitives are required to support the diagnostic techniques in operational deployments. Specifically, primitives concerning model-internal observation (tensor-channel attestation, intermediate-representation logging) may emerge as necessary; primitives concerning institutional process (decision-authority routing, escalation pathway capture) may emerge as necessary. The four primitives identified here are the ones the structural argument supports; the empirical sufficiency question is open.

The institutional-self-knowledge demotion is a structural call we may have gotten wrong. Section 5 demoted institutional self-knowledge from primitive to emergent-property-of-the-four-primitives. The argument was that an institution operating the four primitives well accumulates institutional self-knowledge as a byproduct. This is a strong claim about emergence, and emergence claims are notoriously difficult to verify. If implementation work in Paper 2 reveals that institutional self-knowledge does not emerge robustly from the four primitives — that an architecture can satisfy the four and still fail to produce the durable institutional understanding the strawman material identified as valuable — then institutional self-knowledge needs to be promoted back to primitive status, with its own property-level specification. The demotion is open to revision; the position paper takes it as a structural call but does not foreclose its reversal.

The primitives' operational tractability is empirical and unproven. The paper argues for the primitives at the property level. Whether implementations of these primitives are tractable for community banks under cost and resource constraints is empirical and not addressed by the paper's structural argument. Tamper-evident temporal capture is operationally expensive; evidence binding requires storage and indexing infrastructure; three-dimensional confidence characterization complicates downstream consumption of model outputs; structural drift typology requires multiple monitoring pipelines rather than a single performance dashboard. The aggregate cost of an honest implementation may exceed what community banks can sustainably absorb, in which case the primitives' value is structural-but-aspirational rather than structural-and-deployable. The empirical

question is the work of Paper 2 and of pilot deployments; the position paper's claims rest on the structural argument and stop where the empirical argument begins.

6.5 The meta-uncertainty: theory of effect

The paper's theory of how a position paper produces effect is itself uncertain. Position papers can produce effect through several mechanisms: they can shape regulatory thinking before the regulation is fully developed; they can give examiners interpretive tools they apply within existing authority; they can give architects design vocabulary that propagates through the builder community; they can give the regulated community advance notice of where regulatory practice is heading, allowing institutions to architect ahead of explicit requirements. Each mechanism has different timescales, different audiences, and different success criteria.

The paper's primary intended effect is at the architectural level: shaping what gets built so that built systems are diagnosable in the ways the frame requires. The conclusion's strong claims about regulatory direction support this primary effect; the diagnostic techniques in Section 4, framed as architect-primary, support it. The paper's secondary effects — on examination practice, on institutional anticipation, on regulatory direction — depend on the primary effect succeeding well enough that examiners encounter systems built with the frame in mind, that institutions perceive the regulatory direction the paper articulates as their actual direction, that regulators read the frame as deepening their existing intuitions about FS AI RMF's purpose.

If the primary effect fails — if architects do not adopt the frame and continue building systems without the diagnostic primitives — the secondary effects also degrade. Examiners trained in the techniques apply them to systems that cannot answer the diagnostic questions, producing findings of structural inadequacy that institutions resist on grounds of incommensurability with how systems are actually built. The frame becomes academic rather than operational. The position paper has produced commitments that practice does not honor.

This is the deepest uncertainty the paper carries: whether the structural arguments developed here, however well-reasoned, can produce the practical effect they aspire to. The paper's working assumption is that the regulatory community is sufficiently aligned with the empty-chair frame's underlying intuitions that the frame, articulated and operationalized, will find purchase in practice. The assumption is reasonable but not established. The paper's effect, like the regulatory direction it argues for, is a prediction more than a result.

6.6 What this section earns

We have named uncertainties at every level of the paper's argument. We have not resolved them. The point of the section is not to claim that the uncertainties are minor or that they will be resolved in subsequent work; the point is to mark where the paper's structural arguments stop and where empirical, political, and practical questions begin. The uncertainties are not weaknesses to be hidden; they are the boundary of what a structural argument can establish, and the boundary is where the next work begins.

The position paper's load-bearing function is to articulate structural commitments that the regulatory and architectural communities can adopt or reject, building on what the structural argument establishes and addressing what it leaves open. The uncertainties named here are the open questions; the structural arguments in Sections 1 through 5 are what the paper offers as established. We have tried to mark the boundary clearly, so that what the paper claims can be evaluated separately from what the paper hopes.

7 Conclusion

The Financial Services AI Risk Management Framework arrived in February 2026 with 230 control objectives (U.S. Department of the Treasury 2026) and no interpretive frame for what they collectively protect. Read as a checklist, the framework is operationally daunting and conceptually opaque. Read as a structure, it has an underlying coherence — empty-chair representation — that makes both interpretation and operationalization tractable, and that gives examiners diagnostic leverage the checklist does not provide.

We have argued for three positions in this paper, each of which constrains the others.

The first is that AI governance frameworks in regulated decision contexts are coherent only insofar as they make absent parties audible to the parties present at the moment of decision. The empty chairs are plural and they do not always agree. But the architectural function of regulatory governance is to give them voice in the institutional mechanism, not merely in the institutional documentation. Controls that induce observable, auditable constraints on system behavior or evidence production represent absent parties; controls that assert representation without inducing constraint are decoration. This distinction is not stylistic. It is the difference between governance and governance theater.

The second is that post-hoc explanation is structurally inadequate as a verification regime in adversarial contexts, and bank examination is structurally adversarial. Any architecture that relies on explanation-of-output to verify the reasoning of an AI system in a high-stakes regulated decision has built a Potemkin form of accountability. The empty chair has not been represented; an explanation has been produced and signed. These are not equivalent, and the regulatory apparatus increasingly recognizes that they are not. Frameworks built on the equivalence will not survive the recognition.

The third is that human oversight of AI-generated decisions, in the form currently common in vendor offerings, does not address the verification gap. A human who reviews an explanation has not verified the reasoning. A signature attests to review; it does not attest to verification. The org-chart picture of accountability has been preserved; the empty chair has been moved further from the room, not closer to it. This is the dishonest middle, and its commercial attractiveness will not protect it indefinitely from regulatory scrutiny that learns to ask what an honest verification regime actually requires.

These positions narrow the design space for AI in community banking but do not close it. AI applications in which the verification regime is structurally available — pattern recognition surfacing cases for human investigation, anomaly detection, consistency monitoring across decision histories — remain available and valuable. Applications in which the verification regime is absent require either architectural innovation that closes the gap or honest acknowledgment that the application is not yet ready for deployment in regulated decision contexts. The line between these is not a matter of risk tolerance. It is a matter of whether absent parties are represented or not, and the regulatory apparatus exists precisely to enforce that distinction.

The architectural primitives that empty-chair representation requires — tamper-evident temporal capture, evidence binding, three-dimensional confidence characterization, structural drift typology, diagnostic techniques applied to the attested record — are sketched in this paper at the property level and left to follow-on work for implementation specification. Our concern here has been with the frame, not the build. But the frame's value is in the architecture it forces honest implementations toward, not in any claim that current implementations have arrived. Community banks deploying AI under the FS AI RMF face a genuine question about what *commensurate* compliance means for their context. The frame we have proposed provides the interpretive substrate for answering it. The answer itself remains to be built, and the building is governed by the same structural commitments the frame requires.

The three positions, taken together, name a single structural failure mode the paper has called silence-manufacture: an architecture suppresses access to ground truth, then produces a substitute artifact that occupies the rhetorical position where evidence of ground truth should appear, and reads the substitute as evidence and the suppression as endorsement. Compliance theater is the population-level instance, post-hoc explanation the model-output instance, human-on-explanation the accountability instance. They are not three independent problems; they are three sites where the same structural pattern operates. This is why Goodhart-shaped regulatory reforms — better metrics, harder-to-game targets, more sophisticated controls — fail predictably against the failure mode the paper diagnoses (Goodhart 1984). The pressure has to land at the silence layer, not the metric layer. The architectural primitives sketched in Section 5 are silence-breaking instruments by design; they are not improvements to existing metrics, and reading them through a Goodhart lens will produce frustrated assessments of their adequacy that the frame's actual claim does not warrant.

A final observation, which the frame's logic forces us to make about itself. Silence-producing architectures distort not only the experiences of the parties they suppress but the analytical literatures that try to characterize the phenomenon from outside. Two careful, defensible analyses of the de-banking phenomenon now in the public record — Cato Institute Policy Analysis 1009 (Anthony 2026) reading the phenomenon as governmental debanking driven by Bank Secrecy Act compliance pressure, and unbankedamerica.org's *Atlas of Financial Inclusion* (HR&A Advisors, Inc. 2026) reading it as access barriers driven by fees and ID requirements — describe the same population, draw on the same FDIC numbers (Federal Deposit Insurance Corporation 2024), and reach incompatible causal accounts. Neither analysis is wrong about what it sees. Both are wrong about what the architecture has hidden from them. Reconstruction of the phenomenon's actual causal structure requires breaking the silence the BSA/AML regime legally produces; this is, not coincidentally, what Cato's lead policy recommendation correctly identifies. The literature about an architecture is shaped by the same silence-production that shapes the architecture's effects on the parties it suppresses. The frame predicts this. The de-banking discourse demonstrates it in real time. We name the prediction here so that readers encountering similar partitions in other domains — AI evaluation literature versus deployment-incident reporting, workplace-harassment institutional data versus whistleblower accounts, pharmaceutical-trial publication versus patient-experience reports — can recognize the structural fingerprint and look upstream for the silence-producing architecture rather than treating the partition as an irreducible values dispute.

The empty chairs are watching, and so is the literature about them. The paper's claim is that we should architect for both deliberately, rather than leave them at the mercy of what the architecture has chosen not to make audible.

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